Bank Account Simulation

Using C language to build an efficient bank account system

Group no.
Members:

Tanishk Srivastava	1262253894
Anvi Todkar	1262252159
Tanvi Patil	1262250431
Austy Francis	1262250369
Abhinav Nair	1262253404

Application of C in BANKING

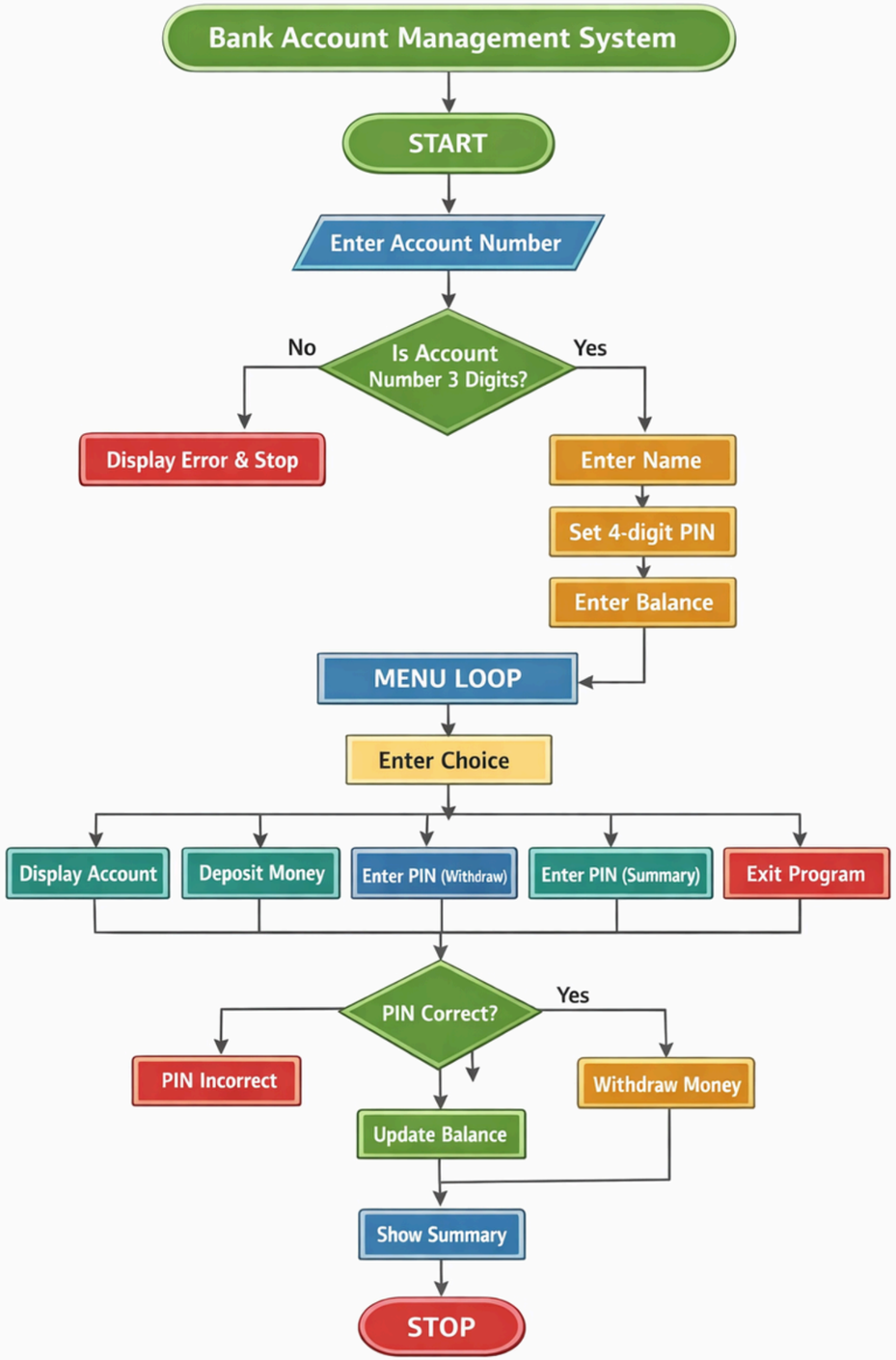
- Structures are used to store account details like account number, name, balance, and PIN.
- printf() and scanf() are used to display messages and take user input.
- if–else conditions are used to check valid account number, correct PIN, and sufficient balance.
- while loop is used to show the menu repeatedly until the user exits.
- switch–case is used to perform different banking operations based on user choice.
- Variables are used to store deposit amount, withdrawal amount, and transaction history.

ALGORITHM

- Start
- Declare structure account with account number, name, balance, and PIN.
- Declare required variables for menu choice, amount, transaction tracking.
- Ask user to enter account number.
- Check if account number is 3 digits:
  - If invalid → display error and stop.
- Ask user to enter name.
- Ask user to set a 4-digit PIN.
- Ask user to enter initial balance.
- Display menu repeatedly using a loop:
  - Display options:
    - Display Account Details
    - Deposit Money
    - Withdraw Money
    - Transaction Summary (PIN required)
- Read user's choice.
- Perform action based on choice:
  - Option 1: Display account details.
  - Option 2:
    - Ask deposit amount.
    - If amount > 0 → update balance and deposit record.
  - Option 3:
    - Ask for PIN.
    - If PIN correct → allow withdrawal if balance sufficient.
  - Option 4:
    - Ask for PIN.
    - If PIN correct → display transaction summary.
  - Option 5: Exit program.
  - Otherwise → display invalid choice.
- Repeat menu until user selects Exit.
- Stop
- Exit

FLOWCHART

Flowchart gives a structured idea of how the program progresses.
This flowchart helps better understand the flow of bank simulation system.



METHODOLOGY

1. Structure Creation
   - A structure named account is used to store account number, name, balance, and PIN.
2. Account Setup
   - The user enters account number, name, PIN, and initial balance.
3. Menu Display
   - A menu is shown repeatedly using a while loop.
   - The user selects options to perform different operations.
4. Display Account Details
   - Displays account number, name, and current balance.
5. Deposit Money
   - The user enters an amount to deposit.
   - The amount is added to the balance if it is valid.
6. Withdraw Money
   - The user must enter the correct PIN.
   - Money is withdrawn only if sufficient balance is available.
7. Transaction History
   - Displays total deposited amount, total withdrawn amount, and last transaction
8. Program Termination
   - The program exits when the user selects the exit option.

INPUT

```
Enter your account number (3 digits): 123
Enter your name: tanishk
Set a 4-digit PIN: 3456
Enter initial balance: 1000
```

```
--- MENU ---
1. Display Account Details
2. Deposit Money
3. Withdraw Money
4. Transaction History
5. Exit
Enter your choice: 3
```

OUTPUT

```
---Account Details---
Account Number : 123
Name           : tanishk
Balance        : 800.00
```

```
4. Transaction History
5. Exit
Enter your choice: 4
Enter PIN: 3840
Incorrect PIN.
```

FUTURE ENHANCEMENTS

1. PIN Attempt Limit: Restrict wrong PIN attempts (example: lock account after 3 tries).

2. Full Transaction History: Store and display multiple deposits and withdrawals instead of only last transaction.

3. Interest Calculation: Add feature to calculate interest on savings balance.

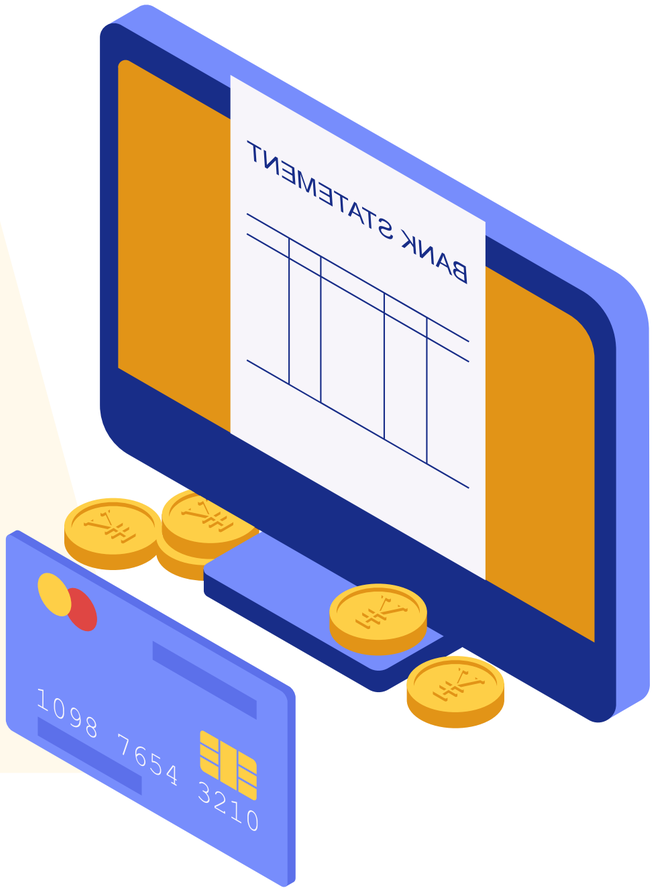
4. Account Types: Provide account types like savings/current with different rules.

5. ATM features: Add options like fast cash, mini statement, and balance enquiry

CONCLUSION

- It demonstrates how real-life banking activities like depositing money, withdrawing money, and maintaining transaction records can be managed in a structured and secure way.

- The project helped us understand how programming concepts are applied to solve practical problems and how logic, accuracy, and validation are important in real systems.



THANKYOU